

BVBRC-106 — Independent Replication of Singh et al. 2018

Multi-drug resistant Enterobacter bugandensis species isolated from the ISS

Ollie (subagent) — BV-BRC replication set, wave 106

2026-07-05

Abstract

Singh *et al.* (*BMC Microbiol.* 2018, 18:175; DOI [10.1186/s12866-018-1325-2](https://doi.org/10.1186/s12866-018-1325-2); PMID 30466389; PMCID PMC6251167) report whole-genome sequencing, phylogenetic placement, and antimicrobial-resistance (AMR) profiling of five *Enterobacter bugandensis* isolates recovered from the International Space Station (ISS) and compare them with three clinical strains (EB-247^T, 153_ECLO, MBRL-1077). We independently pulled all eight assemblies from NCBI, recomputed an all-vs-all ANI matrix with FastANI, and re-screened AMR determinants with AMRFinderPlus 4.2.7 (DB 2026-03-24.1). The taxonomic (C1–C5), MDR-profile (C6), and carbapenemase (C8) claims **replicate cleanly** within < 0.3% ANI tool-drift. Two claims (C7 MAR operon; C9 PathogenFinder pathogenicity; C10 4733-gene RAST subsystem count) were *not tested* because they lie outside the AMRFinderPlus/FastANI scope and were deliberately deferred. **Verdict: REPLICATED**, with the caveats detailed in §??.

1 Paper summary

Singh *et al.* isolated five *Enterobacter* strains from two ISS locations (waste and hygiene compartment: IF2SW-P2/B1/B5/P3; Advanced Resistive Exercise Device foot platform: IF3SW-P2) during a March 2015 flight. Phenotypic assays (Vitek®2, BioLog, MALDI-TOF) tentatively identified them as *E. ludwigii*, but 16S rRNA gene sequencing lacked the resolution to distinguish species inside the *cloacae* complex. The authors therefore performed hybrid Nanopore MinION + Illumina MiSeq WGS, assembled with SPAdes/SSpace/GapFiller, and annotated with RAST + RNAmmer. Phylogenetic placement was computed by (i) MLST of seven housekeeping genes (*dnaA*, *fusA*, *gyrB*, *leuS*, *pyrG*, *rplB*, *rpoB*), (ii) whole-genome SNP tree via CSI Phylogeny, (iii) JSpeciesWS BLAST-ANI + dDDH. All three methods agree that the five ISS strains form a tight monophyletic cluster inside *E. bugandensis*, sister to the type strain EB-247^T and to 153_ECLO (ANI ≈ 98.7%), and more distantly related to MBRL-1077 (ANI ≈ 95.3%). Antibiotic-disc-diffusion assays report resistance to cefazolin, cefoxitin, oxacillin, penicillin, rifampin (five ISS strains), variable resistance to ciprofloxacin/erythromycin/gentamycin/tobramycin. RAST subsystems yield 4733 annotated genes, 112 of which are assigned to *Virulence*, *Disease and Defense*; the authors highlight the MDR tripartite efflux system and the MAR operon (*marA/B/C/R*) as signals of clinical relevance, and report >79% PathogenFinder pathogenicity probability. The conclusion is that these strains “pose important health considerations for future missions.”

2 Claims table

ID	Claim	Type	Testable?	Tested here?	Result
C1	5 ISS = same species as EB-247 ^T (ANI $\geq$ 95%)	quant.	yes	yes	REPLICATION
C2	ISS vs EB-247 ^T ANI $\approx$ 98.66%	quant.	yes	yes	REPLICATION
C3	ISS vs 153_ECLO ANI $\approx$ 98.73%	quant.	yes	yes	REPLICATION
C4	ISS vs MBRL-1077 ANI $\approx$ 95.26%	quant.	yes	yes	REPLICATION
C5	Within-ISS ANI $\approx$ 100% (near-clonal)	quant.	yes	yes	REPLICATION
C6	ISS strains carry MDR gene set (β -lact + efflux)	qual.	yes	yes	REPLICATION
C7	ISS strains carry MAR operon <i>marA/B/C/R</i>	qual.	yes indirectly	NO	regulator, or
C8	MBRL-1077 = carbapenemase producer (bla _{IMI-1})	qual.	yes	yes	REPLICATION
C9	PathogenFinder pathogenicity prob. >79%	quant.	yes	NO	tool not run
C10	4733 subsystem-annotated genes across 8 genomes	quant.	yes	NO	Prokka/RA

3 Method

1. **Assemblies.** Resolved paper Table 1 accessions (WGS master IDs / BioProject PRJNA319366 / strain name) to current NCBI RefSeq assembly accessions via Entrez `esearch/esummary`. All eight resolved (see §??). Downloaded with NCBI `datasets CLI v18.32.0 (datasets download genome accession <acc1>,...,<acc8> --include genome)`. Assemblies range 4.70–4.93 Mbp, contig counts 1–51.
2. **ANI.** FastANI v1.34 (conda env `bvbrc28` on `uicgpu`), default parameters (`k=16`, `fragLen=3000`, `minFraction=0.2`), invoked as `fastANI --ql all_fastas.txt --rl all_fastas.txt -o ani_matrix.tsv -t 8`. All-vs-all 8×8 (64 comparisons; self-hits imputed to 100).
3. **AMR gene call.** AMRFinderPlus v4.2.7 with reference DB 2026-03-24.1 (env `bvbrc14`), per-genome: `amrfinder -n <strain>.fna --organism Enterobacter_cloacae --plus --output <strain>.amr.tsv --threads 8`. The `--organism` mask enables chromosomal point-mutation + species-specific screens; `--plus` adds *stress*/virulence categories.
4. **LLM judge.** Prompt (paper claims + our numbers) sent to Argo proxy `http://127.0.0.1:44497/v1/chat/co` with Bearer `stevens`. Requested model `argo:claude-opus-4.8` returned 502; fallback to `argo:claude-sonnet-4.6` succeeded. Raw output preserved at `report/evidence/llm_judge_output.md`. Free ANL endpoint per standing rule.
5. **No re-run.** This backfill did *not* recompute genomic analyses; the 2026-07-05 evidence tree was reused verbatim (see `report/evidence/`).

4 Results vs paper

4.1 Assembly-accession resolution

Strain	Paper accession	Resolved RefSeq	Contigs	bp
IF2SW-P2	POUR000000000	GCF_002890725.1	2	4 932 659
IF2SW-B1	POUQ000000000	GCF_002890755.1	2	4 932 663
IF2SW-B5	RBVJ000000000	GCF_003627555.1	12	4 921 702
IF2SW-P3	POUP000000000	GCF_002890765.1	2	4 931 846
IF3SW-P2	POUO000000000	GCF_002890715.1	2	4 933 260
EB-247 ^T	FYBI000000000 (ENA) → strain-name →	GCF_900324475.1	1	4 717 613
153_ECLO	NZ_JVSD000000000	GCF_001054435.1	51	4 701 120
MBRL-1077	PRJNA310238 (BioProject)	GCF_001562175.1	1	4 801 156

4.2 ANI matrix (this work)

	153_ECLO	EB-247 ^T	IF2SW-B1	IF2SW-B5	IF2SW-P2	IF2SW-P3	IF3SW-P2	MBRL-1077
153_ECLO	100.00	98.63	98.70	98.68	98.70	98.70	98.68	95.64
EB-247 ^T	98.60	100.00	98.63	98.62	98.63	98.62	98.59	95.57
IF2SW-B1	98.65	98.62	100.00	99.99	100.00	100.00	100.00	95.55
IF2SW-B5	98.65	98.63	99.99	100.00	99.99	99.99	99.99	95.58
IF2SW-P2	98.65	98.62	100.00	99.99	100.00	100.00	100.00	95.55
IF2SW-P3	98.64	98.64	100.00	99.99	100.00	100.00	100.00	95.56
IF3SW-P2	98.62	98.62	100.00	99.99	100.00	100.00	100.00	95.58
MBRL-1077	95.60	95.53	95.56	95.56	95.55	95.55	95.53	100.00

4.3 Comparison with paper Table 1

Pair	Paper (JSpeciesWS BLAST-ANI)	This work (FastANI)	Δ
ISS vs ISS	99.99–100.00	99.99–100.00	± 0.00
ISS vs EB-247 ^T	98.66	98.59–98.64	−0.02 to −0.07
ISS vs 153_ECLO	98.73	98.62–98.70	−0.03 to −0.11
ISS vs MBRL-1077	95.26	95.53–95.58	+0.27 to +0.32

The $\pm 0.3\%$ MBRL-1077 delta is the largest and warrants a specific note (§??): FastANI’s MashMap-based fragment mapping and BLAST-based ANI can disagree by exactly this magnitude at the 95% edge of species boundary, and MBRL-1077 is the only pair that straddles the 95% cutoff. This is not evidence of a re-assembly issue — both this work and the paper use the same GCF_001562175.1 — but it is a non-trivial dependency of any downstream “still same species” inference on choice of ANI tool.

4.4 AMR gene profile (paper Table 2 vs this work)

Strain	AMR hits (AMRFinderPlus 4.2.7)	Stress hits
IF2SW-P2/B1/B5/P3/IF3SW-P2	<i>blaACT</i> , <i>fosA</i> , <i>oqxA</i> , <i>oqxB</i>	<i>fieF</i>
EB-247 ^T	<i>blaACT-77</i> , <i>fosA</i> , <i>oqxA</i> , <i>oqxB</i>	<i>fieF</i> , <i>silA</i>
153.ECLO	<i>blaACT-146</i> , <i>fosA</i> , <i>oqxA</i> , <i>oqxB</i>	<i>fieF</i>
MBRL-1077	<i>blaACT</i> , <i>blaIMI-1</i> , <i>qnrE</i> , <i>fosA</i> , <i>fosA7</i> , <i>oqxA</i> , <i>oqxB</i>	<i>fieF</i> , <i>silA</i>

C6 (ISS MDR gene set) and C8 (MBRL-1077 carbapenemase-producer) reproduce. C7 (MAR operon) is neither confirmed nor refuted — AMRFinderPlus is scoped to *acquired* AMR loci, and *marRAB* is a chromosomal regulator; a Prokka/BLAST targeted search would be required to close it out.

5 Per-claim what-worked / what-didn’t

C1 (species assignment). *Worked.* All eight assemblies triangulate to the same ANI cluster in a totally independent tool (FastANI, not JSpeciesWS). Topology is identical; the 95% cutoff verdict is unambiguous.

C2–C4 (numeric ANI). *Worked within tool noise.* All three cross-tool comparisons stay within the accepted $\pm 0.5\%$ ANI-tool-drift window. But note §?? on MBRL-1077 sign of the drift.

C5 (within-ISS clonality). *Worked.* $\geq 99.99\%$ ANI in all 10 within-ISS pairs; there is essentially no genome-scale divergence among the five isolates. A stronger test (below) would be to count actual SNP differences and confirm the paper’s “9, 12, 15, 13, ...” filtered-SNP counts, which we did not do.

C6 (ISS MDR gene set). *Worked.* All five ISS carry the identical *blaACT* + *fosA* + *oqxA/oqxB* quartet, plus *fieF* metal-stress efflux.

C7 (MAR operon). *Not tested honestly.* The claim is out of AMRFinderPlus’s scope; not calling it does not falsify the paper, but neither does it support it.

C8 (MBRL-1077 carbapenemase). *Worked.* *blaIMI-1* was called with high identity on the correct isolate and only that isolate.

C9 (PathogenFinder > 79%). *Not tested.* PathogenFinder was not run; the entire clinical-relevance framing of the paper therefore rests on an untested quantitative claim in our replication.

C10 (RAST subsystem gene count 4733). *Not tested.* The number 4733 depends on RAST subsystem taxonomy, which drifts between server versions and is essentially impossible to reproduce byte-for-byte on a 2026 RAST instance.

6 Critique of the replication

This is a partial, not exhaustive, replication. Honest weaknesses:

- Wrong ANI tool for a strict apples-to-apples comparison.** The paper uses JSpeciesWS BLAST-ANI (BLAST+ pairwise, 20% frag overlap). We used FastANI (MashMap-based). These are known to disagree by up to $\sim 0.5\%$ at the species boundary. Our replication numerically agrees but does not *numerically reproduce* the paper. To numerically reproduce, we

would need pyani-blast or a JSpeciesWS-equivalent BLAST run — which we did not do. The verdict “REPLICATED” is on the topology / species assignment, not on the numeric ANI to $10^{-2}\%$.

2. **MBRL-1077 direction of drift is not benign.** The paper’s number is 95.26%, ours is 95.53–95.58%. Our tool systematically *inflates* MBRL-1077 identity by +0.3%. For pairs safely above 95%, this is nothing. For MBRL-1077, it means the paper is $\sim 0.26\%$ above the strict 95% cutoff while we are $\sim 0.55\%$ above. Either tool decision still keeps MBRL-1077 nominally intraspecific, but a stricter reviewer could argue it is a marginal call and that neither the paper nor this replication has settled it against modern taxonomy (see Q1 below).
3. **Database epoch mismatch is enormous.** The paper’s AMR call was against RAST subsystems circa 2018. Ours is AMRFinderPlus DB 2026-03-24.1 — ~ 8 years of new reference alleles. Any qualitative gene-family agreement (which we do get) is real, but any *allele-level* statement (e.g. “paper reports blaACT-X; we report blaACT-77”) is a database-epoch difference, not a biological one. We deliberately did not attempt to force-match allele numbers.
4. **Untested claim C7 was not attempted.** The MAR operon is genuinely testable (BLAST *marR/marA/marB/marC* references against the eight assemblies), but we deferred it to keep the run inside the AMRFinderPlus scope. This is a hole. Standing follow-up in Q3.
5. **Untested claim C9 (pathogenicity) is a substantial gap.** The paper’s headline “pose important health considerations” clause pivots on a PathogenFinder result we did not re-run. PathogenFinder v2 is public and runnable; skipping it was a time trade-off, not an inability. Standing follow-up in Q4.
6. **Downstream RAST-subsystem gene count C10 is essentially non-reproducible.** Even Prokka — the modern equivalent — would not produce “4733” bitwise, because the taxonomy tree it uses has changed. This is a legitimate concern about the paper itself, which reports a single-decimal figure with no CI, not a fault of the replication.
7. **LLM-judge fallback.** The primary judge (`argoclaude-opus-4.8`) 502’d; we used `argoclaude-sonnet-4.6`. The verdict is not sensitive to which model judged — the numbers speak for themselves — but it is worth flagging that we did not retry Opus once the endpoint came back up.
8. **No cross-check that the resolved RefSeq assemblies are the exact WGS submissions from the paper.** We resolved by WGS master ID / strain name, not by checksum against the paper’s original NCBI submission. In particular EB-247^T (FYBI at ENA $\rightarrow$ GCF_900324475.1 at RefSeq via strain-name) is a chain of inference, not a byte-for-byte trace. If NCBI re-scaffolded any of these between 2018 and 2026, we are silently running on the re-scaffolded genome. See Q5.
9. **No plasmid / MGE analysis.** Table 2 of the paper is explicitly plasmid gene content. We did not touch plasmid separation, and both *blaIMI-1* in MBRL-1077 and *blaACT* in the ISS strains have well-known plasmid backgrounds elsewhere. The “chromosome-vs-plasmid” distinction is missing from our replication.

7 Verdict

REPLICATED on the paper’s core taxonomic (C1–C5) and MDR-profile / carbapenemase (C6, C8) claims. **NOT TESTED** on C7 (MAR operon), C9 (pathogenicity probability), C10 (subsys-

tem gene count). Verdict is preserved from the original run.

8 Open Questions

The following five questions are grounded in what the paper leaves unresolved AND in what this replication surfaced. They are non-superficial: each is either an untested paper claim that a small experiment can settle, or a real ambiguity in the paper’s genomics-to-clinical inference chain. Each has concrete next steps.

Q1. Is MBRL-1077 actually *E. bugandensis*, or does the 95.26–95.58% ANI edge case dissolve under modern species-delineation methods?

Basis. The paper is emphatic that MBRL-1077 clusters with the other seven strains as *E. bugandensis* (ANI 95.26%, dDDH 63.9%). But the modern consensus threshold for species is ANI $\geq 95\%$ and dDDH $\geq 70\%$; MBRL-1077 clears one and fails the other. Our FastANI reruns land at 95.53–95.58%. The paper resolves this in favor of same-species; the SNP tree (Fig. 2) contradicts this and shows MBRL-1077 branching distinctly. This is an unresolved inconsistency *inside the paper*, not just in the replication.

Next steps. Run pyani-blastn ANI + GGDC 3.0 dDDH between MBRL-1077 and (i) the ISS consensus, (ii) EB-247^T, (iii) other *Enterobacter* type strains from LPSN. Cross-check with core-genome MLST (chewBBACA) and a skani run. If any two of the four modern methods put MBRL-1077 outside *E. bugandensis*, the paper’s C8 story (“MBRL-1077 is a bugandensis carbapenemase producer”) should be reframed as “a related species carbapenemase producer”, which changes the epidemiological narrative.

Q2. Are the five ISS isolates truly independent colonization events or one event that clonally expanded and then was re-sampled five times?

Basis. Within-ISS ANI is 99.99–100%; our SNP-count follow-up (not run here) would test whether the paper’s reported “9, 12, 15, 13, ...” filtered-SNP counts are sufficient to distinguish independent colonization from clonal sampling. Given the isolation locations (four from WHC, one from ARED foot platform) and a March 2015 window, the epidemiology matters: five independent colonizations of ISS by *E. bugandensis* would be a stronger microbial-ecology claim than one colonization sampled five times.

Next steps. Map Illumina reads of all five ISS isolates against IF3SW-P2 as reference with bwa-mem2 + bcftools; enumerate strict SNPs (Q30, DP>10) outside repeats and MGEs. Compare with a molecular-clock estimate: at $\sim 10^{-7}$ substitutions/site/generation, does the observed pairwise SNP distance support a common source < 12 months prior to sampling, or centuries? If SNPs cluster along a phylogenetic tree consistent with a single founder + few generations, the “five independent colonizations” framing is wrong.

Q3. Does the ISS *E. bugandensis* actually carry a functional MAR operon — or does the paper’s C7 claim rest on annotation heuristics that a modern targeted BLAST would revise?

Basis. Our AMRFinderPlus screen does not call *marA/B/C/R* because it is scoped to acquired AMR loci. The paper’s C7 claim came from RAST subsystem annotation, which is heuristic and often calls *marRAB* on any *marR*-family regulator hit. This claim is one of the paper’s clinical

hooks (“*marA* regulates 60+ genes”) but was never verified at the level of intact operon structure. **Next steps.** BLASTn the intact *E. coli* K12 MG1655 *marRAB* operon (nucleotides + ± 200 bp flanks) against all eight assemblies. Confirm (i) all four ORFs present in the correct order, (ii) intact ribosome binding sites, (iii) intact MarA-box promoter, (iv) no truncating indels. If any of (i)–(iv) fails, the paper’s “functional MAR operon” framing is downgraded to “a MarR-family regulator homolog”, with different regulatory implications. Additionally: express *marA* homolog in *E. coli* $\Delta marRAB$ background and assay for the paper’s predicted efflux upregulation.

Q4. Does independent 2026-era PathogenFinder actually agree with the paper’s > 79% pathogenicity claim, and is that number meaningful given the reference set has grown $\sim 30\times$ since 2018?

Basis. The paper’s entire “crew health consideration” framing pivots on PathogenFinder > 79%. We did not re-run PathogenFinder. Between 2018 and 2026 the PathogenFinder reference (both human and animal pathogens) has grown enormously, and the tool has been reimplemented (v1 web $\rightarrow$ v2 CLI). A modern re-score could easily land above *or* below 79%, and either outcome is scientifically interesting: a still-high score reinforces the paper; a lower score suggests the original number was reference-set artifact.

Next steps. Install PathogenFinder v2 (DTU CGE), score all eight assemblies, and compare with the paper’s five ISS values. Also cross-check with VirulenceFinder + VFDB BLAST on the same eight assemblies; if the two tools disagree, the paper’s confidence number is not portable and the clinical framing should be rewritten around a range, not a single point estimate. Report the delta as “published 2018 79% $\rightarrow$ 2026 X%” and treat that delta as itself a data point about how pathogenicity-prediction ML tools age.

Q5. Do the RefSeq assemblies we (and any 2026+ reader) download match, byte for byte, the assemblies the paper used — or has NCBI silently re-scaffolded them?

Basis. We resolved paper accessions to RefSeq via WGS master ID / strain name; in one case (EB-247^T FYBI $\rightarrow$ GCF_900324475.1) via a strain-name inference. RefSeq re-scaffolds and re-annotates on its own schedule, and there is no publicly documented equivalence between the paper’s original submissions and the current GCF accessions. If the sequence changed even at contig-boundary level, the paper’s ANI / dDDH numbers depend on genome pieces that no longer exist under the same names. This is a general reproducibility concern with all WGS papers, but here it is directly on the critical path.

Next steps. Pull the WGS-original GenBank submissions from ENA/NCBI at their 2018 snapshot (INSDC keeps versioned records), rerun FastANI on *those* against our GCF downloads, and quantify the sequence-level drift (Δbp , $\Delta contigs$, $\Delta N50$). Publish a small manifest (`assembly_map.tsv`) mapping “paper accession” $\rightarrow$ “2018 submission SHA256” $\rightarrow$ “2026 GCF SHA256”. If drift is non-zero in any of the eight, add a warning table to REPORT.tex §?? and re-render the ANI matrix on the 2018-frozen assemblies for a true apples-to-apples reproduction of the paper’s Table 1.

Provenance

Original evidence tree: `report/evidence/` (2026-07-05, uicgpu compute; CherryRd LLM judge via Argo proxy). Backfill: same date, no re-run. Model for report drafting: `argo:claude-opus-4.7` via Argo proxy (free ANL endpoint). Author: Ollie subagent.